

Chronobid AI Item Verification Engine - Workflow

Phase 1: Knowledge Base Setup (Offline/Background Task)

1. Fetch Data: The system calls the Met Museum API and requests images for specific categories (e.g., Clocks, Coins, Sculptures, Pottery, Vintage Clothes).
2. Download: The AI downloads a sample of these verified antique images and their details (material, period) and saves them to the 'dataset_images' folder.
3. Generate AI Fingerprints: The OpenAI CLIP model processes every downloaded museum image and converts it into a mathematical vector (an AI fingerprint). These fingerprints are saved locally so the AI has a verified baseline to compare against.

Phase 2: Live Item Verification (User Action)

1. Image Upload: A seller on the Chronobid website uploads an image of an item (e.g., an ancient coin) to create a new auction listing.
2. API Request: The Chronobid website automatically sends this image in the background to the AI Verification API (running on port 8001) via an HTTP POST request.
3. Zero-Shot Classification: The AI engine (using CLIP) instantly looks at the image and predicts its category out of the allowed list with a confidence score (e.g., 'Coin' - 98% confidence).
4. Vector Similarity: The AI generates a mathematical fingerprint for the user's uploaded image and calculates the 'Cosine Similarity' against all the verified Met Museum fingerprints in the database.
5. Finding Matches: The AI finds the top 3 museum objects whose fingerprints most closely match the user's item.

Phase 3: The AI Report (The Output)

1. Generate Report: The AI Engine compiles a JSON report containing:
 - The predicted Category (e.g., 'Coin').
 - The Category Confidence Score.
 - The Estimated Period (inferred from the closest matching museum piece).
 - The Verification Confidence (how closely it visually resembles the real museum piece).
 - Details of the Top 3 similar museum objects (so humans can visually compare them).
2. Display to Users: The Chronobid platform receives this report. Admins can use the Verification Confidence to auto-flag fake items, and Buyers can see the 'AI Verified Category' tag on the auction page, boosting trust

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and bidding engagement.